

Comptia A+ Assignment

[A+ - Troubleshooting and Helpdesk]

Case Study Practical

Work on a scenario involving multiple hardware issues.

Task

❖ **1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.**

➤ Scenario:

- A user reports that their laptop does not power on even when the AC charger is connected. The problem may be related to the power outlet, charger, charging port, battery, or internal hardware.

➤ Step 1: Check the Power Outlet:

- First, connect another working electrical device to the same power outlet to verify that the outlet is supplying power.

➤ What I am checking for:

- Whether the wall socket is working.
- Whether the power switch is ON.
- Whether there is a loose connection or power interruption.

- Step 2: Inspect the Charger and Power Cable:
 - Carefully inspect the laptop charger, adapter, and cable for visible damage. If the adapter has an indicator LED, check whether it is illuminated.
- What I am checking for:
 - Cuts, burns, frayed wires, or bent connectors.
 - Loose connections between the charger and cable.
 - Whether the charger's indicator light is working.
 - Whether the charger appears physically damaged.
- Step 3: Check the Laptop Charging Port:
 - Inspect the laptop's charging port and carefully connect the charger to it.
- What I am checking for:
 - Dust or debris inside the charging port.
 - Bent or damaged pins.
 - A loose charging socket.
 - Whether the charging indicator LED turns on when the charger is connected.
- Step 4: Check the Battery and Power Connection:
 - If the laptop has a removable battery, remove and reinstall it correctly. If possible, test the laptop with the battery removed and the AC adapter connected.
- What I am checking for:
 - Whether the battery is properly seated.
 - Signs of battery damage or swelling.

- Whether the laptop can power on using AC power alone.
- Whether the battery may be causing the power problem.

➤ Step 5: Test with a Known-Good Compatible Charger:

- Use a known-good compatible charger with the correct voltage, connector, and power rating to test the laptop.

➤ What I am checking for:

- If the laptop powers on with the replacement charger, the original charger is likely faulty.
- If the laptop still does not power on, the problem may be related to the charging port, motherboard power circuit, or another internal hardware component.

➤ Expected Result:

- After completing these five steps, the faulty part can be narrowed down to the power outlet, charger, charging port, battery, or internal laptop hardware. If the laptop powers on with a known-good charger, the original charger is likely the cause.

❖ **2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue. Consider checking both hardware connections and basic display settings.**

➤ Scenario:

- The computer appears to be running, but the monitor displays a blank/black screen. The issue may be caused by the monitor, display cable, incorrect input source, GPU connection, or display settings.

➤ Step 1: Check the Monitor Power:

- First, verify that the monitor is switched ON and connected properly to the power outlet.

➤ What I would check:

- Monitor power LED is ON.
- Power cable is firmly connected.
- Power outlet is working.
- Monitor is not showing a “No Signal” or similar message.

➤ Action:

- Reconnect the power cable or try another working power outlet if necessary.

➤ Step 2: Check the Display Cable:

- Check the HDMI, DisplayPort, DVI, or VGA cable connecting the PC to the monitor.

➤ What I would check:

- Cable is firmly connected at both ends.

- Cable or connector is not physically damaged.
- Correct port is being used.
- Action:
 - Disconnect and reconnect the cable. If available, test with another known-good display cable.
- Step 3: Check the Monitor Input Source:
 - Use the monitor's physical buttons/menu to verify that the correct input source is selected.
- For example:
 - HDMI → HDMI
 - DisplayPort → DisplayPort
 - VGA → VGA
- What I would check:
 - The monitor is set to the same input that the computer is connected to.
- Action:
 - Select the correct input source and check whether the display returns.
- Step 4: Check the GPU Connection:
 - If the desktop PC has a dedicated graphics card, turn off the computer and inspect the GPU.
- What I would check:
 - GPU is properly seated in the PCIe slot.
 - GPU power connectors are securely connected.
 - Display cable is connected to the GPU's video output, not the motherboard's video output when a dedicated GPU is being used.

➤ Action:

- Reconnect the GPU/display cable securely and restart the PC.

➤ Step 5: Check Basic Windows Display Settings:

- If Windows is running but the screen is blank or incorrectly configured, press:

➤ Windows + P

- Then select PC screen only or Duplicate.

➤ What I would check:

- Whether Windows is sending the display output to another monitor.
- Whether the display mode is incorrectly configured.

➤ Step 6: Test with Another Monitor or TV:

- Connect the computer to a known-working monitor or TV using a compatible cable.

➤ What I would check:

- If the second display works, the original monitor or its cable may be faulty.
- If the second display also remains blank, the problem may be related to the GPU, computer hardware, or display output.

➤ Step 7: Confirm the Fix:

- Restart the PC and verify that the monitor displays the normal Windows desktop.

➤ I would confirm that:

- Display appears normally.
- No “No Signal” message is shown.

- Correct resolution is displayed.
- The monitor remains stable without flickering or going blank.

➤ Expected Result:

- After completing the troubleshooting steps, the monitor should display the computer's output normally. If the issue persists, the faulty component can be narrowed down to the monitor, cable, GPU, display port, or other PC hardware.

❖ **3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.**

➤ Scenario:

- The wireless keyboard is working normally, but the wireless mouse is not responding. I would troubleshoot the problem systematically to determine whether the fault is with the mouse, USB receiver/port, or Windows settings.

➤ Step 1: Check the Mouse Power:

- First, check the mouse battery and power switch.

➤ What I would check:

- Mouse is switched ON.
- Battery is installed correctly.
- Battery has sufficient charge.

➤ Action:

- Replace the battery with a known-good battery and test the mouse again.

- Step 2: Check the Wireless USB Receiver:
 - Locate the mouse's wireless USB receiver and make sure it is firmly connected to the computer.
- What I would check:
 - Receiver is properly inserted.
 - Receiver is not physically damaged.
 - Mouse and receiver are paired correctly, if pairing is required.
- Action:
 - Remove and reconnect the receiver.
- Step 3: Test Another USB Port:
 - Move the wireless mouse receiver to another known-working USB port.
- What I would check:
 - Whether the mouse works in another USB port.
 - Whether the original USB port is faulty.
- Result:
 - Works in another port → original USB port may have a problem.
 - Doesn't work → continue troubleshooting the mouse or system.
- Step 4: Test the Mouse on Another Computer:
 - Connect the wireless receiver to another working computer and test the mouse.
- What I would check:
 - If the mouse works on another computer, the mouse hardware is probably fine and the original PC may have a USB/driver/settings issue.

- If it does not work on another computer, the mouse, battery, or receiver may be faulty.

➤ Step 5: Check Device Manager:

- On Windows, open:
- Right-click Start → Device Manager → Mice and other pointing devices / Human Interface Devices

➤ What I would check:

- Whether the mouse is detected.
- Any yellow warning/error icons.
- USB devices showing errors.

➤ Action:

- If necessary, update or reinstall the affected device/USB driver and restart the PC.

➤ Step 6: Check Windows Mouse Settings:

- Open:
- Settings → Bluetooth & devices → Mouse

➤ What I would check:

- Mouse is detected by Windows.
- Mouse settings are normal.
- Pointer-related settings have not been incorrectly changed.

❖ **4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one. Think about fans, dust, and thermal paste.**

➤ Scenario:

- A gaming PC is overheating and automatically shutting down during gaming or heavy use. This can happen when the cooling system cannot remove enough heat from the CPU or GPU.

➤ 1. Faulty or Non-Working Cooling Fan:

- The CPU or GPU cooling fan may not be spinning properly, may be blocked, or may have failed.

➤ How I Would Check:

- Shut down the PC and disconnect the power.
- Open the computer case carefully.
- Inspect the CPU and GPU fans.
- Power on the system and check whether the fans spin normally.
- Check for unusual noise, wobbling, or a fan that stops unexpectedly.

➤ Expected Finding:

- If a fan is not spinning or is making abnormal noise, it may need cleaning or replacement.

➤ 2. Excessive Dust Blocking Airflow:

- Dust buildup on the fans, heatsinks, filters, and air vents can restrict airflow and cause the system to overheat.

➤ How I Would Check:

- Turn off and unplug the PC.
- Inspect the CPU heatsink, GPU heatsink, case fans, and air filters.
- Look for thick layers of dust blocking the cooling fins.
- Check whether the case intake and exhaust vents are blocked.
- Expected Finding:
 - Heavy dust buildup can prevent cool air from entering and hot air from leaving the case.
- Action:
 - Clean the dust carefully using appropriate PC-cleaning equipment and ensure that airflow paths are clear.
- 3. Dried or Improperly Applied Thermal Paste:
 - The thermal paste between the CPU and heatsink may have dried out or been applied incorrectly, reducing heat transfer from the CPU to the heatsink.
- How I Would Check:
 - Shut down and unplug the PC.
 - Carefully remove the CPU cooler according to the manufacturer's procedure.
 - Inspect the thermal paste between the CPU and heatsink.
 - Check whether the paste is dried, cracked, or unevenly distributed.
- Expected Finding:
 - If the thermal paste has dried out or is no longer making proper contact, the CPU may transfer heat poorly to the cooler.
- Action:

- Clean the old thermal paste properly and apply a suitable new layer before reinstalling the cooler.